MECHANICAL &
DESIGN ENGINEER

Dana Okonkwo

Lisbon, Portugal · dana@example.org · +351 900 000 000
danaokonkwo.example · EU citizen, no sponsorship

Mechanical engineer working on single-use medical devices from first sketch to tooling release; EU patent holder. Spent three years on the clean-room side of a contract manufacturer before moving into design.

EXPERIENCE

Design Engineer

HYDRA MEDICAL · MAR 2023 – PRESENT · LISBON

- Redesigned the dose counter of a high-volume inhaler (400,000 units a year) with five parts instead of nine.
- Built the drop and cycle test rig used for design verification, wrote the test protocols, and qualified the moulding suppliers behind them.
- Qualified moulding suppliers in Portugal and Poland, including first-article inspection and the PPAP paperwork.
- Ran the weekly DFM reviews with the toolmaker through three rounds of T1 samples, closing 41 of 44 issues before tooling release.
- Built the tolerance stack for the counter mechanism and caught a 0.3 mm interference before the first tool was cut.
- Kept the design history file for the counter: 140 controlled documents through two design reviews.
- Planned and ran formative usability studies (14 patients, three rounds) under IEC 62366, and fed the findings into the design history file.

Prototyping Technician

POLYTECHNIC PROTOTYPING LAB · SEP 2021 – FEB 2023 · PORTO

- Ran a seven-machine prototyping shop — four FDM printers, a resin printer, a benchtop mill and a laser cutter — including the calibration and the repair log.
- Taught the machine-safety induction to about 120 students a year and wrote the handbook for it.
- Designed and printed over 60 jigs and fixtures for student capstone projects, from drill guides to a vacuum-forming frame.
- Kept the shop's machines running: logged and fixed about 300 faults a year, most of them within a day.
- Wrote the lab's equipment grant application: €48,000 for a five-axis mill, approved on first submission.

Freelance CAD Designer

SELF-EMPLOYED · JAN 2021 – AUG 2021 · REMOTE

- CAD and rendering for two small furniture studios, including the assembly drawings they shipped with the product.
- Printed and finished 25 hinge and joint prototypes for the studios' test assemblies.
- Built a searchable parts library for one studio so their joiners could find hardware by fixing type and load.

Clean-room Technician

MEDISURE CONTRACT MANUFACTURING · JUN 2018 – DEC 2020 · PORTO

- Ran ISO 7 clean-room assembly for three catheter products, about 9,000 units a week across two shifts.

- Cut scrap on the balloon-bonding line from 6% to 2.5% by redesigning the bonding fixture.
- Set up SPC charts on the two highest-scrap stations and trained the line leads to read them.
- Ran the validation build for a new catheter line: installation, operational and performance qualification on 11 stations.
- Wrote or rewrote 22 work instructions after a customer audit, all accepted without findings at the follow-up.

Design Intern

ATELIER NORTE · JUN 2017 – SEP 2017 · PORTO

- Modelled and rendered an eight-piece outdoor furniture range for the studio's trade-fair catalogue.

SKILLS

CAD & 3D
SolidWorks (CSWP), Creo, Fusion 360, KeyShot

Fabrication
3D printing FDM/SLA, CNC milling, laser cutting, silicone tooling, injection-moulding DFM

Test & QA
ISO 13485, IEC 62366, design verification, first-article inspection, PPAP

Software
Python, MATLAB, Git, Excel

CERTIFIED

SolidWorks CSWP · **GD&T ASME Y14.5** · **ISO 13485 internal auditor**

PATENT

Dose-counting mechanism for a metered inhaler
CO-INVENTOR · EP 3 000 000 B1 · 2025

EDUCATION

M.Eng. Mechanical Engineering Universidade do Porto
JUL 2021 · DISTINCTION

LANGUAGES

Portuguese	Native
English	C2
Hungarian	B1

INTERESTS

Dinghy sailing · **Bookbinding** · **Restoring mechanical watches**